Injury Severity in Fatal Single-Vehicle Crashes: The Role of Restraint Use and Vehicle Size

Haiyu Xiao[1, *], Qize Zhang[2]
[1]Georgetown University, Washington D.C., 20057, USA, hx111@georgetown.edu
[2]Yale University, New Haven, 06520, USA, qize.zhang@yale.edu
[*]*Corresponding author.*

Abstract*——In traffic accidents, there is a much higher risk of fatal or serious injury for those not wearing seat belts. However, can seat belts reduce the possibility of getting a fatal or serious injury for front seat passengers in a particularly dangerous environment — specifically, when their driver dies in a single-vehicle accident? To study this question, this research uses data from the Fatality Analysis Reporting System (FARS), which contains detailed information on all people involved in fatal crashes occurring on U.S. public roads. For this study, we focused on right front seat passengers aged 18–65 involved in single-vehicle crashes where the driver was killed, analyzing over 3,000 observations across three recent years. Exploratory data analysis, including tables and figures are used to assess how injury severity distributions vary by restraint use and vehicle size. Then, the study fitted linear regression models — with and without interaction terms — to quantify the relationship between restraint use, vehicle size, and injury severity. Results show that seat belt use is strongly associated with reduced injury severity, and that this protective effect is more pronounced in heavier vehicles. The interaction model further suggests that while being unbelted in- creases injury severity in both light and heavy vehicles, the relative benefit of being belted is greater in heavier vehicles. These findings highlight the critical importance of seat belt use in mitigating injury severity, particularly in severe crash scenarios.

***Keywords*——Fatality Analysis Reporting System (FARS), Seat Belt Use, Injury Severity, Vehicle Size, Traffic Safety**

I. INTRODUCTION

In the United States, motor vehicle crashes are a leading cause of death, accounting for approximately 1.38% of all deaths in 2023 [1-2]. During fatal accidents, seat belts are one of the most important safety features, as they keep occupants in place and greatly mitigate injuries during a crash. In NHTSA's studies on motor vehicle accidents, unrestrained passengers consistently demonstrate a higher average death rate compared to those who are properly restrained with seat belts [3]. Crandall et al. [4] also found that during motor vehicle accidents in the United States, the use of lap-shoulder belts was associated with a 72% reduction in fatalities and serious injuries.

To study fatal crashes in greater detail, researchers often rely on the Fatality Analysis Re- porting System (FARS), a nationwide census maintained by the National Highway Traffic Safety Administration (NHTSA). FARS includes every U.S. crash on public roadways resulting in at least one fatality, and because of the limitation to fatal cases, it is particularly valuable for understanding injury severity and survival outcomes among occupants [5]. Prior studies using FARS have consistently shown the protective effects of seat belts and other restraint systems. For example, Levitt and Porter [6] cautioned that because FARS includes only fatal crashes, the absolute effect of restraints may be understated, yet seat belt use remains strongly associated with lower mortality risk. Similarly, Carpenter and Stehr [7] found that primary enforcement seat belt laws reduced youth fatalities by about 8% and serious injuries by 9%.

Beyond evaluating restraint effectiveness, FARS has been applied in a variety of analytical frameworks. Recently, Sulle et al. [8] applied explainable machine learning methods, such as XGBoost with SHAP values, to pedestrian fatality prediction, revealing that factors like age, alcohol or drug impairment, and poor visibility are critical contributors. Despite its strengths, FARS has known limitations. Vaez and Laflamme [9] examined demographic completeness, noting improvements over time but persistent gaps in certain variables. These considerations highlight the importance of interpreting FARS analyses with awareness of potential reporting biases.

However, most research highlights the critical role of seat belts for passengers in general car accident scenarios, with little to no focus on their specific exposure in terms of crash severity, seat position, and vehicle size. In this study, we examine the associations between restraint use and injury severity received across 2 different vehicle size, specifically for passengers aged 18 to 65, seated at the front right in a single vehicle accident severe enough to result in the driver's (front left passenger) death. From 2021 to 2023, more than 280,000 entries of people involved in fatal crashes were recorded, along with over 100 coded data elements that characterize the crashes - including the number of people and vehicles involved, seat positioning, and restrain usage [10]. Using these elements, 3523 observations of front right- seat passengers in single-vehicle accidents where the driver was fatally injured were extracted.

In this paper, section 2 outlines the specific steps of extracting and cleaning the front right passenger data set used in the study, from 3 years of FARS data of all people involved in fatal accidents.

Section 3 includes the exploration and visualization of the data. It is revealed that 71.4% of unbelted passengers and 67.1% of those with seat belts reported unknown experienced fatal or serious injuries, compared to 52.4% of belted passengers. Further visualizations show that in heavy vehicles the seat belts play a greater role in reducing the injury severity. In heavier cars such as SUVs, fatal or serious injuries occurred in 70% of unbelted passengers and 45% of belted passengers; whereas in lighter cars like sedans, these injuries occurred in 73% of unbelted passengers and 59% of belted passengers.

In section 4, to evaluate how restraint use and vehicle size relate to injury severity, two linear regression models are fitted using injury severity as a numeric outcome ranging from 0 (no injury) to 4 (fatal injury). The first model included restraint use and vehicle size as independent predictors. Results showed that unbelted occupants and those with unknown restraint status had significantly higher injury severity scores than belted individuals, and occupants in light vehicles experienced higher severity than those in heavier vehicles. To further explore whether the effect of restraint use differs by vehicle type, we introduced an interaction term in the second model. The interaction model revealed that while unbelted and unknown restraint status consistently increased injury severity, the magnitude of this effect was smaller in light vehicles than in heavy vehicles.

Finally, section 5 presents conclusions drawn based on the models and explorations of the data, along with recommendations and ideas for future research.

II. DATA

A. Data Acquisition

This study utilizes individual-level data from the 2021-2023 release of the FARS dataset. We specifically analyze the person-level file (person.csv), where each observation represents a single individual involved in a fatal crash. The raw dataset contains approximately 280,000 observations and 126 variables, covering a wide range of information, including injury severity, restraint use, vehicle characteristics, and crash time. For this study, we focused on motor vehicle occupants aged 18 to 65, restricting the sample to passengers and drivers involved in crashes with clearly reported injury severity and restraint status.

To ensure independence, we selected the data only for the front right passenger, limiting the sample to crashes involving a single car with at least one passenger in addition to the driver, given that the driver is fatally injured. This refinement helps us to ensure that all the passengers in the sample experienced similar exposure to a fatal crash event. The final working dataset (front_right_passengers.csv) includes 3,483 individuals.

B. Exploratory Data Analysis

When a single-car accident results in the death of the driver, the passenger in the front right seat is exposed to varying levels of injury risk. To study the association between their seat belt usage, the car size, and their injury severity, exploratory analysis and data visualization were performed on the cleaned dataset.

Data pre-processing step includes extracting and cleaning the three main variables used in the study: InjurySeverity, Restraintuse, and CarSize.

The primary outcome variable, InjurySeverity, is an ordinal indicator of injury severity received by the passenger. Based on the original recordings ("Fatal Injury (K)", "Suspected Serious Injury (A)", "Suspected Minor Injury (B)", "Possible Injury (C)", "No Apparent Injury (O)"), the cleaned InjurySeverity is categorized into five levels: Fatal, Serious, Minor, Possible, No Injury.

The main predictor of interest, Restraintuse, indicates seat belt usage of the passenger during the accident. From the FARS data set recordings, categories "Shoulder and Lap Belt Used", "Lap Belt Only Used", "Shoulder Belt Only Used", "Restraint Used - Type Unknown", and "Racing-Style Harness Used" are categorized into "Belted", "Not Reported", "Other", "Reported as Unknown" are categorized into "Unknown", and "None Used/Not Applicable" is converted to "Unbelted".

CarSize categorizes the vehicle in the accident into light and heavy, based on their car type. "Coupe", "Hatchback/Liftback/Notchback", "Convertible/Cabriolet", "Off-road Vehicle - Recreational Off-Road Vehicle (ROV)", and "Sedan/Saloon" are categorized into Light Vehicles, where "Crossover Utility Vehicle (CUV)", "Pickup", "Sport Utility Vehicle (SUV)/Multi-Purpose Vehicle (MPV)", "Sport Utility Truck (SUT)", "Truck", and "Van" are converted into Heavy Vehicles.

These variables were selected to address the research question regarding the association between restraint use, vehicle size, and injury severity in fatal motor vehicle crashes.

Table 1 presents a frequency table generated by grouping by car size and restraint use, showing the percentage of each injury severity level for belted, unbelted, and unknown.

Table 1: Distribution of Injury Severity by Car Size and Restraint Use

CarSize	RestraintUse	InjurySeverity	n	Percentage
Heavy Vehicle	Belted	Fatal	113	15.56
Heavy Vehicle	Belted	Serious	218	30.03
Heavy Vehicle	Belted	Minor	226	31.13
Heavy Vehicle	Belted	Possible	103	14.19
Heavy Vehicle	Belted	No Injury	66	9.09
Heavy Vehicle	Unbelted	Fatal	220	31.47
Heavy Vehicle	Unbelted	Serious	268	38.34
Heavy Vehicle	Unbelted	Minor	141	20.17
Heavy Vehicle	Unbelted	Possible	41	5.87
Heavy Vehicle	Unbelted	No Injury	29	4.15
Heavy Vehicle	Unknown	Fatal	61	41.5
Heavy Vehicle	Unknown	Serious	37	25.17
Heavy Vehicle	Unknown	Minor	25	17.01
Heavy Vehicle	Unknown	Possible	16	10.88
Heavy Vehicle	Unknown	No Injury	8	5.44
Light Vehicle	Belted	Fatal	206	25.69
Light Vehicle	Belted	Serious	268	33.42
Light Vehicle	Belted	Minor	202	25.19
Light Vehicle	Belted	Possible	76	9.48
Light Vehicle	Belted	No Injury	50	6.23
Light Vehicle	Unbelted	Fatal	233	34.42
Light Vehicle	Unbelted	Serious	262	38.7

Light Vehicle	Unbelted	Minor	123	18.17
Light Vehicle	Unbelted	Possible	36	5.32
Light Vehicle	Unbelted	No Injury	23	3.39
Light Vehicle	Unknown	Fatal	91	38.89
Light Vehicle	Unknown	Serious	66	28.21
Light Vehicle	Unknown	Minor	53	22.65
Light Vehicle	Unknown	Possible	13	5.56
Light Vehicle	Unknown	No Injury	11	4.7

Figure 1 is a stacked bar plot that visualizes the relationship between restraint use and injury severity. The height of the colored segments represents the proportion of individuals within that category who experienced each level of injury severity.

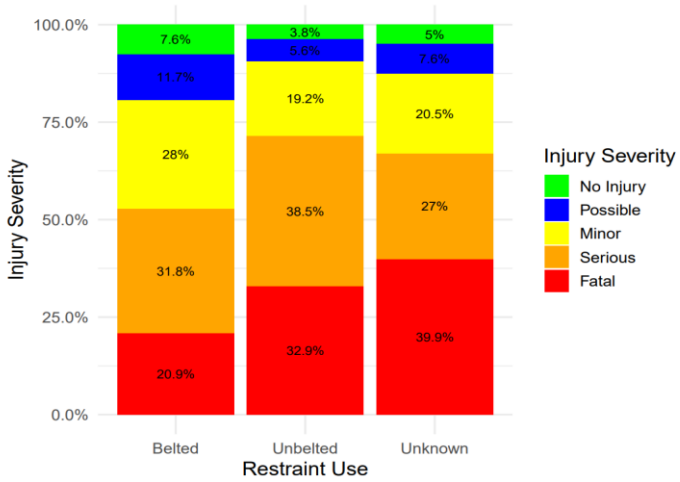

Figure 1: Injury Severity Distribution by Restraint Use for All Car Type

Shown in the Table 1 and figure 1, the distribution of injury severity varies notably by restraint use. Among belted passengers, the most common outcomes were serious injuries (31.8%) and minor injuries (28.0%), while fatalities accounted for 20.9% of cases. In contrast, unbelted individuals experienced a significantly higher fatality rate of 32.9%, and a greater share of serious injuries (38.5%), with much lower proportions of no injury (3.8%) and possible injury (5.6%). For individuals with unknown restraint status, fatalities were the most common outcome at 39.9%, followed by serious injuries (27.0%). These patterns suggest a strong association between restraint use and reduced injury severity, with belted individuals experiencing fewer fatal and serious outcomes compared to those unbelted or with unknown restraint use.

The size of the vehicle — categorized as light or heavy — is another important factor in the study. By limiting the analysis to one vehicle size at a time, tables and visualizations below illustrate the association between restraint use and injury severity for the front seat passenger.

Figure 2 is a stacked bar plot that visualizes the relationship between restraint use and injury severity, for the front seat passenger in heavier cars.

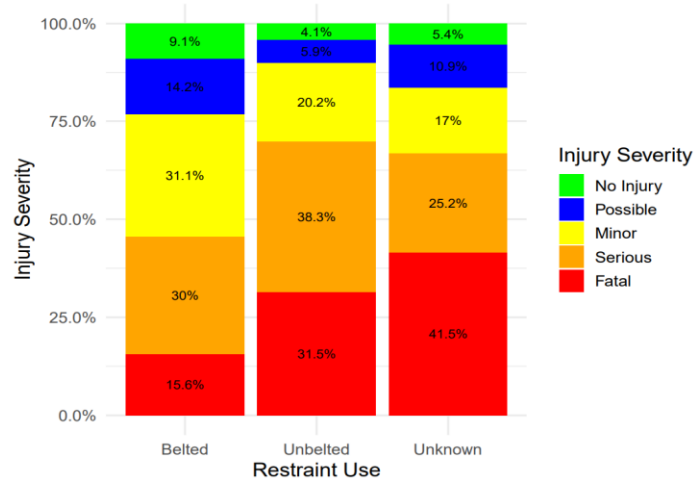

Figure 2: Injury Severity Distribution by Restraint Use in Heavier Cars

In Table 1 and Figure 2, among belted individuals in heavier vehicles, serious (30.0%) and minor (31.1%) injuries are most common, while fatalities account for 15.6% of cases. In contrast, unbelted individuals in heavy vehicles face significantly higher fatality rates (31.5%) and serious injuries (38.3%), with much lower rates of no injury (4.2%) and possible injury (5.9%).

Figure 3 is a stacked bar plot that visualizes the relationship between restraint use and injury severity, for the front seat passenger in lighter cars.

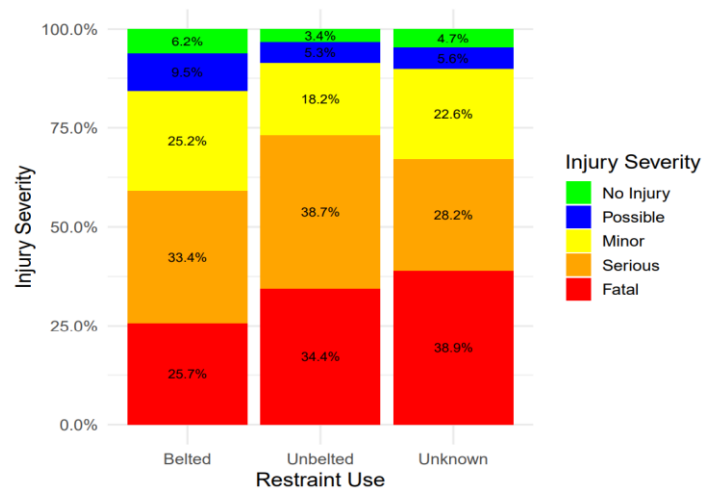

Figure 3: Injury Severity Distribution by Restraint Use in Lighter Cars

In Table 1 and Figure 3, similar distribution is observed for lighter vehicles. Here, the injury severity profile is generally worse. Belted occupants of light vehicles show a higher fatality rate (25.7%) compared to their counterparts in heavy vehicles (15.6%), and slightly more serious injuries (33.4%). Unbelted individuals in light vehicles exhibit the highest fatality rate across all groups (34.4%) and serious injury rate (38.7%), with even lower percentages of minor or no injuries compared to unbelted occupants in heavier vehicles.

Using the tables and the visualizations, it is shown that restraint use consistently correlates with lower injury severity across both vehicle types, but the protective effect appears more pronounced in heavier vehicles. Belted occupants in heavy vehicles show notably lower fatality and serious injury rates than belted occupants in light vehicles. Similarly, while unbelted individuals fare poorly in both vehicle types, their injury outcomes are slightly worse in lighter vehicles. The outcome suggests that both restraint use and vehicle size play significant roles in mitigating injury severity during fatal crashes.

III. EXPERIMENTS

A. Experiment Models

To further investigate the observed relationship between restraint use, vehicle size, and injury severity, linear regression models are fitted, including an interaction term between restraint use and vehicle type. While the frequency tables revealed clear patterns—such as higher injury severity among unbelted occupants and worse outcomes in lighter vehicles — they do not account for potential interactions between these factors or provide a way to quantify their combined effects. A regression model will allow to formally test whether the impact of restraint use on injury severity differs by vehicle size, while also estimating the direction and magnitude of these effects. By modeling injury severity as a numeric outcome and including both main effects and their interaction, the study will evaluate not only the individual contributions of restraint use and vehicle type but also whether their effects vary depending on one another.

In the models, the outcome variable, InjurySeverityVal, is an ordinal numeric encoding of injury severity, where higher values represent more severe outcomes (e.g., 0 = No Injury, 4 = Fatal Injury). The predictors included RestraintUse (categorized as "Belted", "Unbelted", or "Unknown") and CarSize (categorized as "Light Vehicle" or "Heavy Vehicle"), along with their interaction.

Model 1 included restraint use and vehicle size as independent predictors, where its coefficients are shown in Table 2.

Table 2: Coefficients of Linear Model 1 (InjurySeverityVal ~ RestraintUse + CarSize)

term	estimate	std.error	statistic	p.value
(Intercept)	2.362	0.035	67.614	0
RestraintUse Unbelted	0.452	0.041	10.962	0
RestraintUse Unknown	0.408	0.064	6.415	0
CarSize Light Vehicle	0.199	0.039	5.132	0

To further investigate whether the effect of restraint use on injury severity varies by vehicle size, we fitted a linear regression model with an interaction term between restraint use and vehicle type. Table 3 shows the coefficients of model 2:

Table 3: Coefficients of Linear Model 1 (InjurySeverityVal ~ RestraintUse * CarSize)

term	estimate	std.error	statistic	p.value
(Intercept)	2.288	0.041	55.671	0
RestraintUse Unbelted	0.583	0.059	9.942	0
RestraintUse Unknown	0.576	0.1	5.752	0
CarSize Light Vehicle	0.341	0.057	6.004	0
RestraintUse Unbelted: CarSize Light Vehicle	-0.258	0.082	-3.128	0.002
RestraintUse Unknown: CarSize Light Vehicle	-0.294	0.13	-2.27	0.023

B. Experiment Results

Model 1 revealed that, compared to being in the belted group in heavy vehicles (predicted injury severity = 2.36), unbelted individuals in heavy vehicles had an average injury severity score that was 0.45 points higher (predicted = 2.81), while those with unknown restraint status had a score 0.41 points higher (predicted = 2.77) — both differences were statistically significant ($p < 0.001$). Additionally, belted occupants in light vehicles had a predicted score of 2.56, representing a 0.20-point higher injury severity on average compared to belted occupants in heavy vehicles. Unbelted and unknown-status occupants in light vehicles had predicted scores of 3.01 and 2.97, respectively.

Table 4: Predicted Injury Severity by Restraint Use and Vehicle Size (Model 1)

RestraintUse	CarSize	Pred
Belted	Heavy Vehicle	2.36
Unbelted	Heavy Vehicle	2.81
Unknown	Heavy Vehicle	2.77
Belted	Light Vehicle	2.56
Unbelted	Light Vehicle	3.01
Unknown	Light Vehicle	2.97

In model 2, as setting "Belted" and "Heavy Vehicle" as the reference categories, the estimated injury severity value is approximately 2.29. Being unbelted in heavy vehicles significantly increases predicted injury severity to about 2.29 + 0.58 = 2.87, and a similar effect is observed to the "Unknown" seat belt usage group (2.29 + 0.58 = 2.87).

For the belted and light Vehicle group, there is a significant increase of 0.34 in the estimated injury severity comparing to belted and heavy vehicle group (2.29 + 0.34 = 2.63), which again supports the finding that belted occupants in heavy vehicles experience notably lower fatality and serious injury rates than belted occupants in light vehicles. The significant negative interaction terms for the unbelted (–0.26) and

unknown (–0.29) groups in light vehicles reveal an effect: while unbelted and unknown restraint status consistently elevate injury severity, the magnitude of this additional elevation (relative to being belted) is smaller when occupants are in light vehicles compared to heavy vehicles. Specifically, the estimated scores are 2.29 + 0.58 + 0.34 – 0.26 = 2.95 for unbelted occupants in light vehicles, and 2.29 + 0.58 + 0.34 – 0.29 = 2.92 for those with unknown restraint status. This does not imply that light vehicles are safer, but rather that the relative benefit of being belted is greater in heavier vehicles.

Table 5: Predicted Injury Severity by Restraint Use and Vehicle Size (Model 2, with interaction)

RestraintUse	CarSize	Pred
Belted	Heavy Vehicle	2.29
Unbelted	Heavy Vehicle	2.87
Unknown	Heavy Vehicle	2.86
Belted	Light Vehicle	2.63
Unbelted	Light Vehicle	2.95
Unknown	Light Vehicle	2.91

IV. Discussion & Conclusion

To address the research question of whether seat belts reduce the possibility of getting a fatal or serious injury for front seat passengers in a single-vehicle accident where the driver dies, this study used data from the Fatality Analysis Reporting System (FARS), and conducted exploratory data analysis to identify descriptive patterns, followed by linear regression modeling (with interaction terms) to quantify the effects of restraint use and vehicle type on injury severity. This approach provided both descriptive and statistical evidence to evaluate how these factors influence injury outcomes.

Exploratory data analysis shows that belted individuals experience significantly lower rates of fatal and serious injuries (53%) compared to unbelted (71%) or unknown restraint users (67%). This protective effect of seat belts is evident across both heavier and lighter vehicle types, though the severity of injury tends to be greater in lighter vehicles overall. In particular, unbelted occupants in light vehicles suffer the highest fatality and serious injury rates (73%), comparing to belted in light vehicles (59%); while unbelted passengers in heavy cars also suffer high chance of fatal or serious injuries (70%), belted passengers in heavier vehicles exhibit the most favorable outcomes (46%). These patterns highlight the dual importance of wearing restraints and being in larger vehicles in improving survival and reducing injury severity in fatal crashes.

Statistical modeling supports these findings by quantifying the effects of restraint use and vehicle size on injury severity. The interaction model estimated that belted passengers in heavy vehicles had an average injury severity score of 2.29 (on a 1–5 scale). Being unbelted increased this by 0.58 points in heavy vehicles, while unknown restraint use increased it by 0.58 points as well — both statistically significant. Occupants in light vehicles had a 0.34-point higher score compared to heavy vehicles when belted. Interestingly, the interaction terms indicate that while unbelted status still increases injury severity in light vehicles, the additional increase is 0.26 points smaller compared to heavy vehicles; similarly, the "unknown" restraint group's increase is 0.29 points smaller. This suggests that seat belts provide a particularly strong protective benefit in heavier vehicles. Overall, both the descriptive and inferential analyses reinforce the life-saving value of seat belt use and the role of vehicle size in mitigating injury outcomes.

Despite the strength of the above findings, there are some limitations to consider. For instance, the FARS dataset only includes crash in which one or more fatalities occurred, therefore it doesn't generalize to all motor vehicle crashes in all situation, particularly minor ones. On the other hand, although the model's R-squared value is relatively low (4.5%), this is not unexpected given the complexity of factors influencing injury severity in fatal crashes. Many important contributors — such as impact speed, crash dynamics, and passenger health — are either unmeasured in this study or are factors outside the passenger's control. In contrast, restraint use and vehicle type are controllable or influenceable decisions, and this research aims to highlight their individual contributions to injury severity. Despite explaining only a small portion of the total variation, the significant coefficients in the model underscore the meaningful impact that seat belt use and vehicle size have on injury outcomes — reinforcing their importance as public health and policy levers.

Future studies incorporating additional crash-level and person-level variables could further refine risk prediction. Nevertheless, the results of this study underscore the importance of continued investment in seat belt education, enforcement, and technology-based interventions aimed at improving compliance and reducing preventable injuries on the road.

References

[1] Febres, J.D., García-Herrero, S., Herrera, S., Gutiérrez, J.M., López-García, J.R. and Mariscal, M.A. (2020). Influence of seat-belt use on the severity of injury in traffic accidents. *EUROPEAN Transport Research Review*, [online] 12(1), pp.1–12. doi: https://doi.org/10.1186/s12544-020-0401-5.

[2] National Safety Council (2023). *Introduction - Injury Facts*. [online] Injury Facts. Available at: https://injuryfacts.nsc.org/motor-vehicle/overview/introduction/.

[3] National Highway Traffic Safety Administration (2021). *Traffic safety during the COVID-19 public health emergency: January–June 2021*. [online] U.S. Department of Transportation. Available at: https://www.nhtsa.gov/sites/nhtsa.gov/files/2021-10/Traffic-Safety-During-COVID-19_Jan-June2021-102621-v3-tag.pdf

[4] Crandall, C.S., Olson, L.M. and Sklar, D.P. (2001). Mortality Reduction with Air Bag and Seat Belt Use in Head-on Passenger Car Collisions. *American Journal of Epidemiology*, 153(3), pp.219–224. doi: https://doi.org/10.1093/aje/153.3.219.

[5] Koehler, S.A. and Brown, P.A. (2010). *Forensic epidemiology*. Boca Raton: Crc Press.

[6] Levitt, S.D. and Porter, J. (2001). Sample Selection in the Estimation of Air Bag and Seat Belt Effectiveness. *Review of Economics and Statistics*, 83(4), pp.603–615. doi: https://doi.org/10.1162/003465301753237696.

[7] Carpenter, C.S. and Stehr, M. (2008). The effects of mandatory seatbelt laws on seatbelt use, motor vehicle fatalities, and crash-related injuries

among youths. *Journal of Health Economics*, 27(3), pp.642–662. doi: https://doi.org/10.1016/j.jhealeco.2007.09.010.

[8] Sulle, M., Mwakalonge, J., Comert, G., Siuhi, S. and Gyimah, N.K. (2025). *Unraveling Pedestrian Fatality Patterns: A Comparative Study with Explainable AI*. [online] arXiv.org. Available at: https://arxiv.org/abs/2503.17623 [Accessed 17 Aug. 2025].

[9] Vaez, M. and Laflamme, L. (2005). Impaired driving and motor vehicle crashes among Swedish youth: An investigation into drivers' sociodemographic characteristics. *Accident Analysis & Prevention*, 37(4), pp.605–611. doi: https://doi.org/10.1016/j.aap.2005.03.001.

[10] www.nhtsa.gov. (n.d.). NHTSA File Downloads | NHTSA. [online] Available at: https://www.nhtsa.gov/file-downloads?p=nhtsa/downloads/FARS/